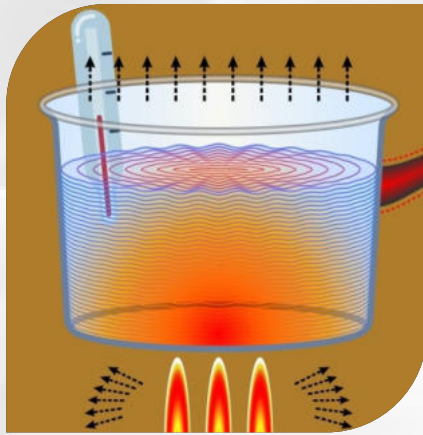


UNIT 09

THERMAL PROPERTIES OF MATTER



HEAT AND TEMPERATURE

"Heat is a form of energy transferred from one body to another as a result of a difference in temperature."



UNIT

The S.I unit of heat is Joule.

FOR EXAMPLE

A hot cup of tea is placed on a table, after some time the tea in the cup becomes cold because surrounding temperature is lower than that of the hot tea. Hence heat flows from hot cup to the surrounding. There are many everyday examples of heat that we experience in our daily lives. Some examples include: Cooking food on a stove or in an oven Feeling the warmth from the sun on a hot summer day Holding a hot cup of coffee or tea Taking a hot shower or bath Turning on a heater on a cold day Touching a hot stove and feeling a burning sensation.

TEMPERATURE

"It is a degree of hotness or coldness of a body. It determines the direction of flow of heat from one body to the other body".

UNIT

The S.I unit of temperature is Kelvin (K).

Heat	Temperature
Heat is a form of energy.	Temperature is a measure of the average kinetic energy of the particles in a substance.
Heat is measured in units of energy like joules or calories.	Temperature is measured in units of degrees like Celsius or Fahrenheit.
Heat is related to the amount of energy in a substance.	Temperature is related to the intensity of the energy.
Heat always flows from hotter objects to cooler objects.	Temperature can be the same between two objects in contact.
	

THERMOMETER

Thermometer is a device used to measure temperature.

FOR EXAMPLE

A clinical thermometer is used to measure the temperature of the human body. Thermometers have different scales to measure temperature. There are three scales of temperature;

1. Celsius ($^{\circ}\text{C}$)

Mostly used for environmental measurements.

2. Fahrenheit ($^{\circ}\text{F}$)

Mostly used for clinical measurements.

3. Kelvin (K)



MOSTLY USED FOR INDUSTRIAL MEASUREMENTS

These three scales of temperature are interconvertible. Therefore, temperature measured in Celsius scale can be converted into Kelvin and Fahrenheit scales as scale.

$$\text{K} = ^{\circ}\text{C} + 273$$

Conversion of temperature from Celsius scale to Fahrenheit scale

$$^{\circ}\text{F} = 1.8^{\circ}\text{C} + 32$$

HEAT CAPACITY AND SPECIFIC HEAT CAPACITY

It is the amount of heat energy required to raise the temperature of an object by a certain degree. It is expressed in units of energy per unit of temperature change, typically in joules per kelvin (J/K) or calorie. Heat capacity is an important concept in thermodynamics, as it determines how much energy must be added or removed from an object in order to change its temperature.

MATHEMATICALLY,

$$C = \frac{Q}{\Delta T}$$

Where **Q** = Amount of heat absorbed and (**ΔT**) is change in temperature. Heat capacity depends upon the nature of material.

FOR EXAMPLE

Two breakers contain equal masses of water and oil is heated by the same gas burner for three minutes. Then it is observed that the temperature of oil may rise twice than water. It is the amount of heat required to raise the temperature of 1 kg of a substance through 10°C is called specific heat capacity of that substance. When comparing the heat capacity of different substances, we are actually talking about their specific heat capacity. Hence specific heat capacity can be defined as:

$$c = \frac{Q}{m\Delta T}$$

Where "**c**" is constant which depends upon the nature of material of the body. This constant is called **specific heat capacity** or **specific heat**. Its SI unit is joule per kilogram per Kelvin (**$\text{J kg}^{-1} \text{K}^{-1}$**).

EFFECT DUE TO LARGE SPECIFIC HEAT OF WATER

We know that water has a large specific heat, due to this quality it plays an important role in everyday life. This property of water has several important effects:

1. The large amount of water in oceans and lakes help to maintain the temperature ranges in their surroundings.
2. Water with coolant is used to reduce the temperature of engine through radiator of vehicle.
3. Water also help to maintain our body temperature.

WORKED EXAMPLE 1

The thermal energy required to raise the temperature of 50g of water from 40°C to 70°C is 6300 Joules. Calculate the specific heat capacity of water.

STEP 1

$$T_1 = 40^\circ\text{C}$$

$$T_2 = 70^\circ\text{C}$$

$$\Delta T = T_2 - T_1 = 70^\circ\text{C} - 40^\circ\text{C} = 30^\circ\text{C} = 30\text{K}$$

$$\Delta Q = 6300\text{ J}$$

$$m = 50\text{ g} = 0.05\text{ kg}$$

$$c = ?$$

STEP 2: Write down the formula and rearrange if necessary

$$c = \frac{\Delta Q}{m\Delta T}$$

STEP 3: Put the values in formula and calculate

$$c = \frac{6300\text{J}}{0.05\text{kg} \times 30\text{K}}$$

$$c = 4200\text{J kg}^{-1}\text{K}^{-1}$$

Hence, specific heat of water is $c = 4200\text{ J kg}^{-1}\text{K}^{-1}$.

LATENT HEAT OF FUSION AND LATENT HEAT OF VAPORIZATION

Latent heat of fusion is the amount of heat energy required to change a substance from solid to liquid state without any change in temperature is called **“heat of fusion”**.

FOR EXAMPLE

The latent heat of fusion of water is 334 J/g or 80 Cal/g. This means that to melt 1 gram of ice at 0°C, 334 J of heat energy must be added without any increase in temperature. Similarly, to freeze 1 gram of water at 0°C, 334 J of heat energy must be removed. The concept of latent heat of fusion is important in understanding phase changes and heat transfer in various physical and chemical processes.

LATENT HEAT OF VAPORIZATION

The latent heat of vaporization is the amount of energy required to change a substance from a liquid state to a gaseous state without a change in temperature is called **“Heat of Vaporization”**.

FOR EXAMPLE

When water is heated on a stove, it reaches a temperature of 100°C. At this point, the energy being added to the water is used to break the intermolecular bonds between the water molecules. This results in the water molecules changing state from liquid to gas, which is known as **vaporization**. During this phase change, the energy being added to the water does not raise the temperature of the water any further. Instead, it is used to overcome the intermolecular forces holding the water molecules together and allow them to separate and become water vapor. This energy is known as the latent heat of vaporization, and it is a measure of the energy required to change a unit mass of a substance from liquid to gas at a constant temperature. In the case of water, the latent heat of vaporization is about 2,260 kJ/kg. This means that for every kilogram of water that is boiled, 2,260 kJ of energy is required to convert it from a liquid to a gas. This example of latent heat of vaporization can help students understand the concept of energy being used to change the state of a substance, and how this energy is not reflected in a change in temperature. This experiment has two parts:

1. Conversion of ice into water.
2. Conversion of water into steam

CONVERT ICE (SOLID) INTO WATER LIQUID**EXPERIMENT**

A simple experiment to demonstrate the transformation of ice into water can be done as follows:

MATERIALS

A small plastic cup or container
Ice cubes
Room temperature water
A thermometer
A timer or clock

PROCEDURE

1. Fill the plastic cup about $\frac{2}{3}$ full with ice cubes.
2. Add room temperature water to the cup until it is nearly full.
3. Stir the mixture gently for a few seconds.
4. Insert the thermometer into the mixture and record the initial temperature.
5. Wait for 10–15 minutes, stirring occasionally, and record the temperature at regular intervals (e.g. every minute or every 2 minutes).
6. Observe what happens to the ice cubes as the temperature rises.
7. Keep stirring until all the ice cubes have melted and the temperature has stabilized.

RESULTS

As the ice cubes are added to the room temperature water, the initial temperature of the mixture drops. As time passes, the water absorbs heat energy from the surroundings and the temperature of the mixture gradually rises. The ice cubes start to melt and transform into liquid water. The melting process takes some time and is accompanied by a gradual increase in temperature.

Once all the ice has melted, the temperature of the water stabilizes at a value close to room temperature.

CONCLUSION

This experiment shows that ice can be transformed into water when it is exposed to a higher temperature. The melting process is a gradual one, and it involves the absorption of heat energy by the ice. This experiment can be used to introduce the concept of phase transitions and the role of temperature in changing the physical state of matter.

CONVERT WATER (LIQUID) INTO STEAM (GAS)

EXPERIMENT:

A simple experiment to demonstrate the transformation of water into gas can be done as follows:

CONCLUSION

A stove or hot plate

A pot with a lid

Water

A thermometer

PROCEDURE

1. Fill the pot with water up to two-thirds of its capacity.
2. Place the pot on the stove or hot plate and turn on the heat.
3. Cover the pot with the lid and wait for the water to boil. You will notice steam escaping from the pot.
4. Measure the temperature of the boiling water with a thermometer. It should be around 100°C .
5. Remove the lid and watch as the steam continues to rise from the pot.
6. Hold a cool surface, such as a metal spoon or a mirror, above the pot, close to the steam. The steam will condense on the surface, forming tiny droplets of water.
7. Turn off the heat and let the pot cool down. The remaining water in the pot will slowly evaporate over time, but no steam will be visible without heating it again.

EXPLANATION

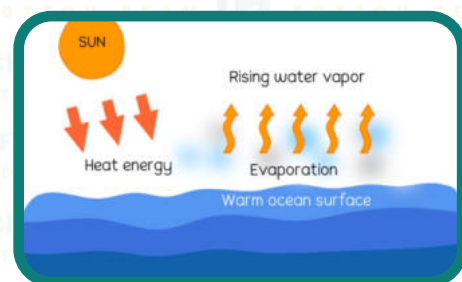
When the water is heated, its molecules gain energy and move faster, eventually breaking the bonds that hold them together as a liquid. At the boiling point, these molecules have enough energy to overcome the atmospheric pressure and escape into the air as water vapor or steam. The temperature of the steam is the same as the boiling water (100°C at sea level) but it is in a gaseous state and takes up much more space. As the steam cools down, its molecules lose energy and move slower, allowing them to rejoin and condense back into liquid water.

EVAPORATION PROCESS

Evaporation is the process by which a liquid substance, such as water, changes into a gas or vapor state.

There are some examples of evaporation process:

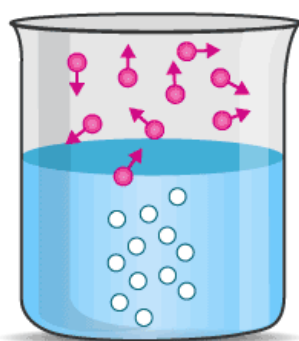
1. Boiling water to make tea or coffee.
2. Water evaporating from a swimming pool.
3. Saltwater evaporation to form salt crystals.
4. Rainwater evaporating from the ground after a rainstorm.



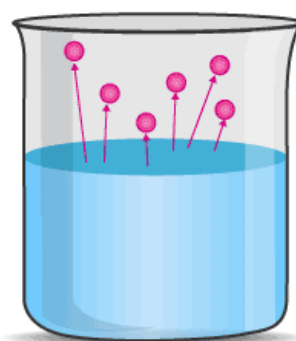
5. Sweat evaporating from our skin to cool us down.
6. Lakes and rivers evaporating during droughts.
7. Gasoline evaporating from a car's fuel tank.
8. Snow and ice evaporating on a sunny day.
9. Ink drying on paper.

DIFFERENCE BETWEEN BOILING POINT AND EVAPORATION

Boiling	Evaporation
Occurs at a specific temperature	Occurs any temperature
Requires sufficient heat to convert liquid to gas	Occurs without adding heat, but can be speed up with heat
Occurs throughout the liquid	Occurs at the surface of the liquid
Can be used for distillation and separation of mixtures	Can be used for drying and cooling
Takes longer than evaporation	Occurs faster than boiling point
Example: Water boils at 100°C	Example: Clothes drying on a clothesline



Boiling



Evaporation

The table below shows the freezing and boiling of some important solvents:

SOLVENT	FREEZING POINT(°C)	BOILING POINT (°C)
Water	0.0	100
Acetic acid	17.0	118.1
Benzene	5.5	80.2
Ethanol	-114.7	78.4
Chloroform	- 63.5	61.2

EVAPORATION CAUSES COOLING

When evaporation occurs, the molecules of water with greater Kinetic energy from its surface. This is because when a liquid evaporates, it takes energy from the surrounding environment to overcome the intermolecular forces holding the liquid together. This energy is usually in the form of heat, which is absorbed by the evaporating molecules, leading to a decrease in the temperature of the remaining liquid and the surrounding environment. That's why we feel cool when we sweat, because the sweat evaporates, taking heat away from our bodies. Some liquids have low boiling points due to which they change from liquid to vapor easily at ordinary temperature, these liquids are called **"volatile liquids"**.

FOR EXAMPLE

An example of evaporation causing cooling is the process of sweating in humans. When we sweat, the moisture on our skin evaporates into the air, taking heat with it and causing a cooling effect on our skin. This is because the heat energy required for evaporation is taken from the surrounding environment, including our skin, causing the surface to cool down. This is why sweating is an effective way for our bodies to regulate our internal temperature, particularly when we're hot and need to cool down. Another example is when a wet cloth is left in the sun. As the water on the cloth evaporates, it takes heat energy from the cloth, causing it to cool down. This is why wet clothes or towels can be used to provide a quick, temporary cooling effect on the body, particularly in hot weather or after exercise.

FACTORS THAT INFLUENCE SURFACE EVAPORATION

Surface evaporation is the process by which a liquid transforms into a gas or vapor and enters

the atmosphere. This process is influenced by several factors, including:

- 1. TEMPERATURE:** Higher temperatures increase the rate of evaporation, as more heat energy is available to break the intermolecular bonds and convert liquid into gas.
- 2. HUMIDITY:** The higher the humidity in the surrounding air, the slower the rate of evaporation, because the air is already saturated with water vapor.
- 3. WIND SPEED:** Increased wind speed enhances the rate of evaporation, as it moves the moist air away from the surface, maintaining a drier environment.
- 4. SURFACE AREA OF LIQUID:** Evaporation rate is also affected by the surface area of the liquid exposed to the atmosphere. Larger surface areas promote higher evaporation rates.

5. PRESSURE: Changes in atmospheric pressure can affect the rate of evaporation, particularly at high altitudes where the pressure is lower.

6. TYPE OF LIQUID: The physical properties of the liquid such as density, viscosity, and surface tension can affect the rate of evaporation.

7. SOLAR RADIATION: The amount of solar radiation received by the surface can impact the rate of evaporation as it provides energy to the surface and heats the liquid.

8. DEPTH OF THE LIQUID: The depth of the liquid also affects the rate of evaporation. Shallow water bodies will evaporate more quickly than deep ones because there is less water to evaporate.

These factors interact with each other to determine the rate of evaporation for a given liquid under specific conditions.

THERMAL EXPANSION

Thermal expansion is the tendency of matter to change its shape, area, volume, and density in response to a change in temperature. In general, when matter is heated, its molecules gain kinetic energy and move faster, leading to an increase in volume. There are three main types of thermal expansion:



1. LINEAR EXPANSION: This occurs when a material expands in only one dimension, usually length.

2. AREA EXPANSION: This occurs when a material expands in two dimensions, usually length and width.

3. VOLUME EXPANSION: This occurs when a material expands in all three dimensions, usually length, width, and height.

EXAMPLES OF THERMAL EXPANSION

A TRAIN TRACK EXPANDS DUE TO HIGH TEMPERATURE: When the temperature rises, the metal tracks used in trains tend to expand, which can cause the tracks to buckle or warp, which can lead to accidents. To counteract this, gaps are left between the tracks to allow for expansion.

EXPANSION OF BRIDGES AND HIGHWAYS: Concrete and steel bridges and highways also expand when exposed to high temperatures, causing them to bulge and warp. Engineers have to account for thermal expansion and contraction when designing these structures.

EXPANSION OF METAL PIPES: When hot water is passed through metal pipes, they tend to expand. This can lead to damage to the pipes if not properly installed with expansion joints. **Expansion of Buildings:** As the temperature rises, buildings made of materials such as concrete, brick, and steel tend to expand. This can cause cracks to form in walls and floors.

EXPANSION OF CAR TIRES: During hot summer days, car tires tend to expand, increasing the pressure inside the tire. This can cause the tire to burst if not properly inflated.

EXPANSION OF ALUMINUMS FOIL: When heated, aluminum foil expands, making it useful in cooking and baking as it can be moulded to fit different shapes and sizes of containers.

EXPANSION OF WATER: Water also expands when heated, which is why water heaters have expansion tanks to accommodate the increased volume of water. This can prevent damage to the pipes and other components of the system.

EXPANSION OF SOLID: Expansion of solids is the phenomenon where the size, shape, and volume of a solid material change due to the influence of various factors such as heat, pressure, and stress.

LINEAR EXPANSION: The Expansion in length of Solid object on heating is called linear expansion. It is one dimensional expansion as it occurs only along the length of the object. Suppose a rod of some material with original length L_1 at initial temperature T_1 . When heated the rod and its temperature become T_2 and length becomes L_2 . Therefore

T_1 = initial temperature

L_1 = initial length

L_2 = final Length

T_2 = final temperature

Change in Temperature = $\Delta T = T_2 - T_1$ ----- (1)

Change in Length = $\Delta L = L_2 - L_1$ ----- (2)

It has been experimentally proved that change in length is directly proportional to the original length and change in temperature. Therefore

$$\Delta L = \alpha L_1 \Delta T$$

Where ΔL is the change in length of the object, α is the coefficient of linear expansion, L is the original length of the object, and ΔT is the change in temperature.

VOLUMETRIC EXPANSION

Volumetric expansion is the change in volume of a material due to a change in temperature. When an object is heated, its particles gain kinetic energy and move faster, causing the material to expand. Similarly, when an object is cooled, its particles lose kinetic energy and move slower, causing the material to contract. The amount of volumetric expansion or contraction can be calculated using the coefficient of volume expansion (β), which is a property of the material. The formula for calculating volumetric expansion is:

$$\Delta V = V_1 \beta \Delta T$$

Where:

ΔV = Change in Volume

V_1 = Initial Volume

β = Coefficient of Volume Expansion

ΔT = Change in Temperature

The coefficient of volume expansion varies for different materials and can be found in tables of material properties. It is typically measured in units of $(1/K)$, where K represents degrees Kelvin, the unit for temperature



NOTE

As linear expansion occurs in one dimension, whereas volume expansion occurs in three dimensions. Hence, the coefficient of volume expansion " β " is three times greater than the coefficient of linear expansion " α ".

Therefore

$$\beta = 3\alpha$$

APPLICATION AND CONSEQUENCES OF THERMAL EXPANSION

Thermal expansion of solids is useful in some situations of daily life and in some situations it creates problems;

APPLICATIONS

Thermal expansion refers to the tendency of materials to expand or contract in response to changes in temperature. This phenomenon has a wide range of applications in various fields. Here are some examples:

1. Bridges and Buildings: Thermal expansion and contraction are important factors to consider in the design and construction of bridges and buildings. Materials such as concrete, steel, and wood can expand or contract significantly with changes in temperature, which can cause stress on the structure. Engineers must take into account the expected temperature ranges in the location of the structure and design accordingly to prevent structural damage.

2. Plumbing Systems: Pipes made of metals such as copper or steel can expand or contract with changes in temperature. This can cause stress on the pipes and fittings, leading to leaks or bursts. To prevent this, plumbers use expansion joints, which allow the pipes to expand and contract without causing damage.

3. Thermal Expansion Valves: These are components used in refrigeration and air conditioning systems to regulate the flow of refrigerant. They use the thermal expansion of a temperature-sensitive liquid or gas to control the flow of refrigerant through the system.

4. Railways: Railway tracks are made of steel, which can expand and contract with temperature changes. This can cause the tracks to buckle or warp, which can be dangerous for trains. To prevent this, railway engineers leave gaps between the sections of the tracks to allow for expansion and contraction.

5. Cooking: Thermal expansion is also used in cooking. For example, when bread is baked, the dough expands as it heats up, creating air pockets that make the bread rise. Similarly, when water is heated, it expands, which can cause it to boil and create steam. These are just a few examples of the many ways in which thermal expansion is used in various applications.

REAL AND APPARENT EXPANSION OF LIQUIDS

The real expansion of a liquid refers to the actual increase in volume that occurs when the temperature of the liquid increases. When the temperature of a liquid increases, the average kinetic energy of the molecules within the liquid increases, causing the molecules to move farther apart and occupy more space. This results in an increase in the volume of the liquid. The apparent expansion of a liquid, on the other hand, refers to the expansion that is observed when the liquid is viewed from the outside. This expansion is affected not only by the real expansion of the liquid, but also by the expansion or contraction of the container that holds the liquid. For example, if a liquid is contained in a glass flask, and the flask expands or contracts due to a change in temperature, the apparent expansion of the liquid will be affected by the change in the volume of the flask.

Be aware of your own worth, use all of your power to achieve it. Create an ocean from a dewdrop. Do not beg for light from the moon, obtain it from the spark within you.

Allama Iqbal

